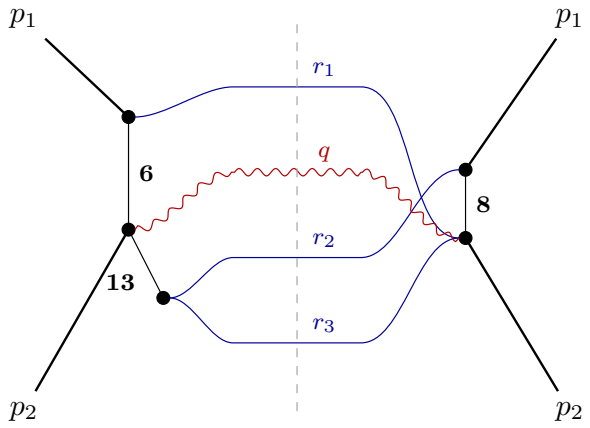
Top sectors as cut diagrams

Amplitude (left) × conjugate (right) for $p_1 p_2 \to V(q)\, r_1 r_2 r_3$, glued along the cut (dashed): r_1, r_2, r_3 (blue), q (red, $q^2 = m^2$). The rapidity condition $-q \cdot p_1 + u\, q \cdot p_2 = 0$ of every family is not a line. Internal lines carry their DD number. **Regular propagators** (DD numbers, all legs incoming): $6 = (p_1 + r_1)^2$; $7 = (p_2 + r_1)^2$; $8 = (p_1 + r_2)^2$; $9 = (p_2 + r_2)^2$; $10 = (p_1 + r_3)^2$; $11 = (p_2 + r_3)^2$; $12 = (r_1 + r_2)^2$; $13 = (r_2 + r_3)^2$; $14 = (r_1 + r_3)^2$; $15 = (p_1 + r_1 + r_2)^2$; $16 = (p_2 + r_1 + r_2)^2$; $17 = (p_1 + r_1 + r_3)^2$; $18 = (p_2 + r_1 + r_3)^2$; $19 = (p_1 + r_2 + r_3)^2$; $20 = (p_2 + r_2 + r_3)^2$; $21 = (r_1 + r_2 + r_3)^2$; $22 = (p_1 + r_1 + r_2 + r_3)^2$; $23 = (p_2 + r_1 + r_2 + r_3)^2$; $24 = (p_1 + p_2 + r_1)^2$; $25 = (p_1 + p_2 + r_2)^2$; $26 = (p_1 + p_2 + r_3)^2$; $27 = (p_1 + p_2 + r_1 + r_2)^2$; $28 = (p_1 + p_2 + r_1 + r_3)^2$; $29 = (p_1 + p_2 + r_2 + r_3)^2$.



FAM[55] (topo55), sector with 3 of 7 lines

6	$(p_1 + r_1)^2$	amplitude
13	$(r_2 + r_3)^2$	amplitude
8	$(p_1 + r_2)^2$	conjugate

pinched: $7 = (p_2 + r_1)^2$, $14 = (r_1 + r_3)^2$, $18 = (p_2 + r_1 + r_3)^2$, $22 = (p_1 + r_1 + r_2 + r_3)^2$